

Hibernate Setup with NetBeans - Step by Step Guide

1. Java JDK Install

Download and install Java JDK (e.g., JDK 21) from the official Oracle website. Ensure JAVA_HOME is set and 'java -version' works in the terminal.

2. NetBeans Setup

Download Apache NetBeans (latest version). Install and open it.

3. Create Maven Project

In NetBeans: File > New Project > Maven > Java Application.
Give your project a name like 'Hibernate_Demo' and finish.

4. Add Dependencies in pom.xml

Add the following dependencies inside <dependencies> tag:

```
<dependency>
  <groupId>org.hibernate.orm</groupId>
  <artifactId>hibernate-core</artifactId>
  <version>6.4.4.Final</version>
</dependency>
```

```
<dependency>
  <groupId>mysql</groupId>
  <artifactId>mysql-connector-j</artifactId>
  <version>8.0.33</version>
</dependency>
```

5. Create Resource Folder

Right-click on 'src/main' > New > Folder > Name it 'resources'.
Mark this folder as Resources Folder if required.

6. Add hibernate.cfg.xml in resources folder

hibernate.cfg.xml example content:

```
<?xml version='1.0' encoding='utf-8'?>
<!DOCTYPE hibernate-configuration PUBLIC ... >
```

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```
<hibernate-configuration>
<session-factory>
  <property name='hibernate.connection.url'>jdbc:mysql://localhost:3306/hibernatedb</property>
  <property name='hibernate.connection.username'>root</property>
  <property name='hibernate.connection.password'>your_password</property>
  <property name='hibernate.dialect'>org.hibernate.dialect.MySQLDialect</property>
  <mapping class='com.mycompany.hibernate_demo.User'/>
</session-factory>
</hibernate-configuration>
```

7. Create Entity Class (User.java)

In package com.mycompany.hibernate_demo create User.java with annotations:

```
@Entity
public class User {
    @Id
    private int id;
    private String name;
    private String email;
}
```

8. Create Hibernate_Demo.java (Main class)

In same package, create main class to:

- Open Hibernate session
- Save User object to DB
- Close session

9. Run the Project

Right-click project > Run.

Hibernate will connect to DB, create table, insert data.

10. Output & Confirmation

Check NetBeans output window. You'll see logs:

- Table created using DDL
- Insert statement executed
- Connection opened & closed
- Message: User saved successfully!